Phase 1

What is Programming?

Programming is the process of writing instructions that tell a computer what to do. These instructions are written using programming languages.

Just as people use languages like English or Hindi to communicate with each other, programmers use programming languages to communicate with computers and solve problems.

Example:

```
print("Hello, World!")
```

The computer reads this instruction and displays the message on the screen.

What is Python?

Python is a popular programming language known for its simple and easy-to-read syntax. It is designed to be beginner-friendly, allowing new programmers to learn coding more easily.

Because Python code often looks similar to plain English, it is widely used in education and industry.

Example

```
>> name = "Kuwar"
>> print(name)
```

Output:

Kuwar

Features of Python

1. Easy to Learn

Python has a clean and simple syntax, making it easier to understand than many other programming languages.

2. Free and Open Source

Python can be downloaded and used without any cost. Its source code is publicly available for anyone to view and improve.

3. High-Level Language

Python allows programmers to focus on solving problems rather than dealing with complex system-level details.

4. Cross-Platform Support

Python programs can run on different operating systems such as Windows, Linux, and macOS.

5. Large Community

Python has a huge community of developers who create tutorials, tools, and libraries that make development easier.

6. Wide Range of Applications

Python is used in many areas, including:

- Web Development
- Data Analysis
- Machine Learning
- Artificial Intelligence
- Automation
- Cybersecurity
- Software Development

Installing Python

Python can be downloaded from the official website:

Python.org

Installation Steps

1. Visit the Python website.
2. Download the latest version of Python.
3. Run the installer.
4. Make sure to select **"Add Python to PATH"** during installation.
5. Complete the installation process.

Check Installation

Open Command Prompt or Terminal and run:

```
>> python --version
```

If Python is installed correctly, the version number will be displayed.

Example

Python 3.13.5

Your First Python Program

```
>> print("Hello World")
```

Output:

Hello World

Explanation

- print() is a built-in Python function.
- It displays information on the screen.
- "Hello World" is a string (text).

Quick Revision

✓ Programming = Giving instructions to a computer.

✓ Programming Language = A language used to communicate with computers.

✓ Python = Simple, powerful, beginner-friendly programming language.

✓ print() = Used to display output.

✓ Python is used for Web Development, AI, ML, Automation, and Data Science.

Phase 2 – Variables

Variables are one of the most important concepts in Python. Almost every Python program uses variables.

1. What is a Variable?

A variable is a container used to store data in memory.

Think of a variable as a labeled box where you can keep information and use it later.

Example

name = "Kuwar"

Here:

- name → Variable name
- "Kuwar" → Value stored in the variable

2. Creating Variables

```
name = "Kuwar"
age = 20
height = 5.8
```

You can print variables using print():

```
print(name)
print(age)
print(height)
```

Output:

```
Kuwar
20
5.8
```

3. Variable Assignment

The = operator assigns a value to a variable.

city = "Patna"

Python reads it as:

Store "Patna" inside the variable city.

4. Rules for Naming Variables

☑ Valid Names

```
name = "Kuwar"
user_age = 20
student1 = "Rahul"
```

✖ Invalid Names

```
1name = "Kuwar"    # Starts with a number
user-age = 20      # Hyphen not allowed
class = "Python"   # Reserved keyword
```

Rules

1. Must start with a letter or _
2. Cannot start with a number
3. Cannot contain spaces
4. Cannot use Python keywords
5. Variable names are case-sensitive

Example:

```
name = "Kuwar"
Name = "Rahul"

print(name)
print(Name)
```

These are two different variables.

5. Multiple Variable Assignment

Assign Different Values

```
name = "Kuwar"
age = 20
city = "Patna"
```

Assign in One Line

```
name, age, city = "Kuwar", 20, "Patna"
```

6. Assign Same Value to Multiple Variables

```
x = y = z = 100
```

```
print(x)
print(y)
print(z)
```

Output:

```
100
100
100
```

7. Updating Variables

Variables can be changed anytime.

```
age = 20
print(age)

age = 21
print(age)
```

Output:

```
20
21
```

The old value is replaced by the new value.

8. Variable Types

Python automatically determines the type of data.

```
name = "Kuwar"   # String
age = 20        # Integer
height = 5.8    # Float
is_student = True # Boolean
```

9. Check Variable Type

Use type() to check the data type.

```
name = "Kuwar"

print(type(name))
```

Output:

```
<class 'str'>
```

More examples:

```
print(type(10))
print(type(5.5))
print(type(True))
```

10. Dynamic Typing

Python allows a variable's type to change.

```
x = 10
print(x)

x = "Hello"
print(x)
```

Output:

```
10
Hello
```

This feature is called **Dynamic Typing**.

11. Constants (By Convention)

Python does not have true constants, but programmers use uppercase names.

```
PI = 3.14159
GRAVITY = 9.8
```

These values are intended not to be changed.

12. Best Practices

Use Meaningful Names

✅ Good

```
student_name = "Kuwar"
total_marks = 450
```

❌ Bad

```
a = "Kuwar"
b = 450
```

Use Snake Case

```
first_name = "Kuwar"
total_marks = 450
```

Avoid:

```
firstName = "Kuwar"
TotalMarks = 450
```

Real-Life Example

```
student_name = "Kuwar"
student_age = 20
course = "Python"

print("Name:", student_name)
print("Age:", student_age)
print("Course:", course)
```

Output:

```
Name: Kuwar
Age: 20
Course: Python
```

Summary

- A variable stores data.
- = is used to assign values.
- Variable names must follow naming rules.
- Python is dynamically typed.
- Use meaningful variable names.
- type() checks the data type of a variable.